

# Detailed Design

Low-Level Design (LLD) - ovvoru module-um epdi work aagum

## Detailed Design na yenna?

Architecture-la pirichcha ovvoru component-um **ullukkulla epdi work aagum** nu detail-a design pannradhu: classes, functions, data flow.

## Yenna Yenna Design Pannanum

- Module list - ovvoru module-oda velai.
- Class design - attributes, methods.
- Function / API design - input, output.
- Data flow - data epdi move aagum.
- Error handling - thappu vandhaa yenna aagum.

## UML Diagrams

| Diagram              | Yenna kaattum                      |
|----------------------|------------------------------------|
| Use Case             | User system-oda yenna pannalam.    |
| Class Diagram        | Classes, avanga relationship.      |
| Sequence Diagram     | Time-order-la messages epdi pogum. |
| Activity / Flowchart | Process step by step flow.         |
| State Diagram        | Object-oda states maaruradhu.      |

## Module Design Template

Module name: \_\_\_\_\_  
Purpose: \_\_\_\_\_  
Inputs: \_\_\_\_\_ Outputs: \_\_\_\_\_  
Depends on: \_\_\_\_\_  
Error cases: \_\_\_\_\_

## API Design Example

| Endpoint    | Method | Input           | Output         |
|-------------|--------|-----------------|----------------|
| /login      | POST   | email, password | token or error |
| /users/{id} | GET    | user id         | user details   |
| /tasks      | POST   | title, date     | created task   |

## Good Design Principles

- Single responsibility - oru module oru velai.
- Low coupling - modules oru-thara-a sarndhu irukka koodadhu.
- High cohesion - oru module-kulla ellam related-a irukkanum.
- DRY - repeat pannaadheenga.